Localising BBH mergers to AGN

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In collaboration with **Fiona Panther**

The origin of binary black holes

- Host galaxy origin of BBHs is uncertain, most common model is stellar collapse of massive binary stars.
- PISN leads to prediction of a "mass gap" (Woosley+ 2007) in the BH mass spectrum.

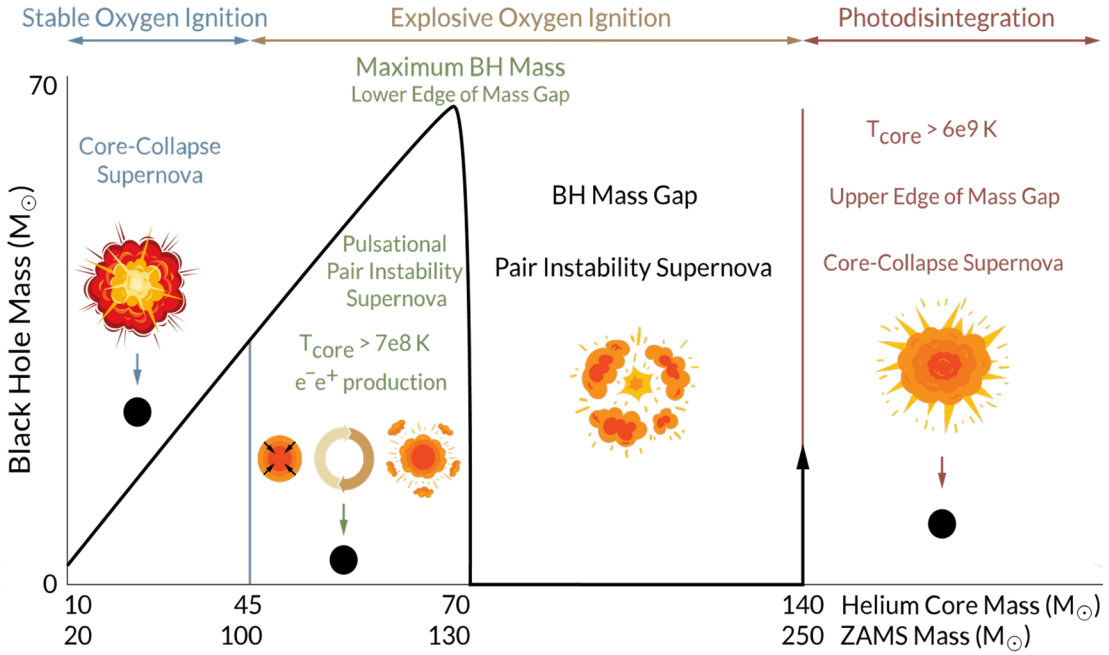
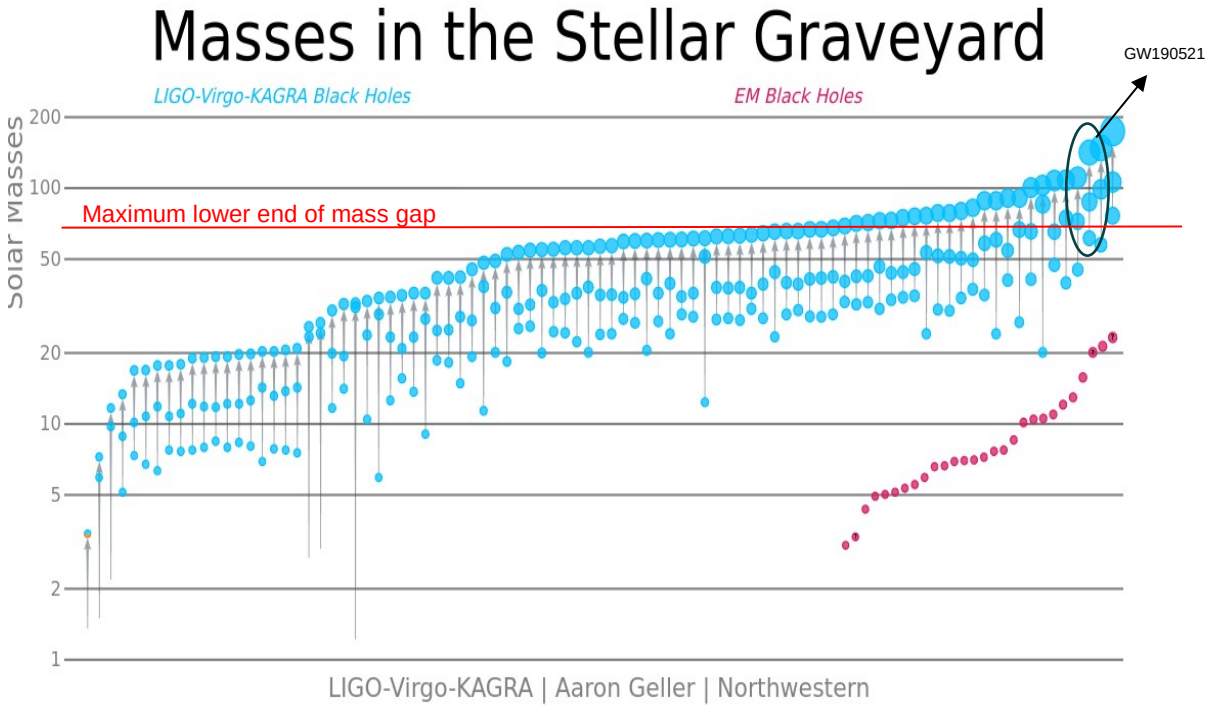


Image credit: Farag+2022

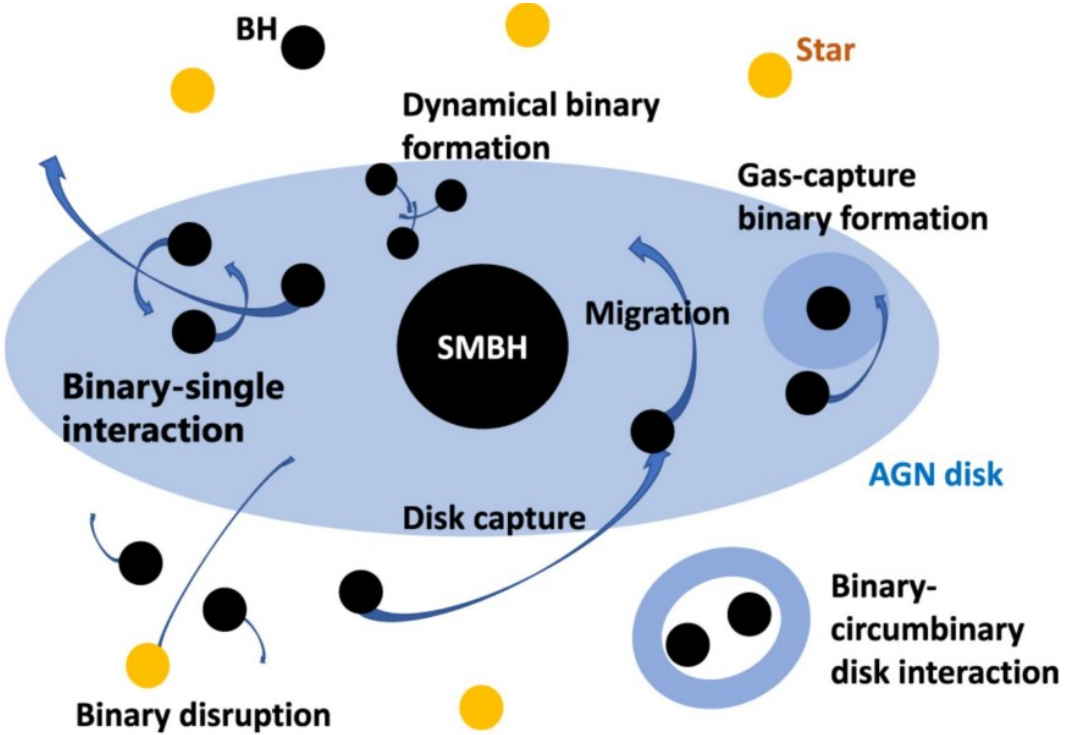
GW190521

- Predicted mass gap does not appear to exist in observed GW detections (Abbott+ 2023)
- GW190521 was the result of the merger of "intermediate mass" BHs (in the mass gap).
- Suggests that either
    1. Stellar evolution models are wrong.
    2. BBHs are not (always) formed from field binaries.



Binaries in AGN

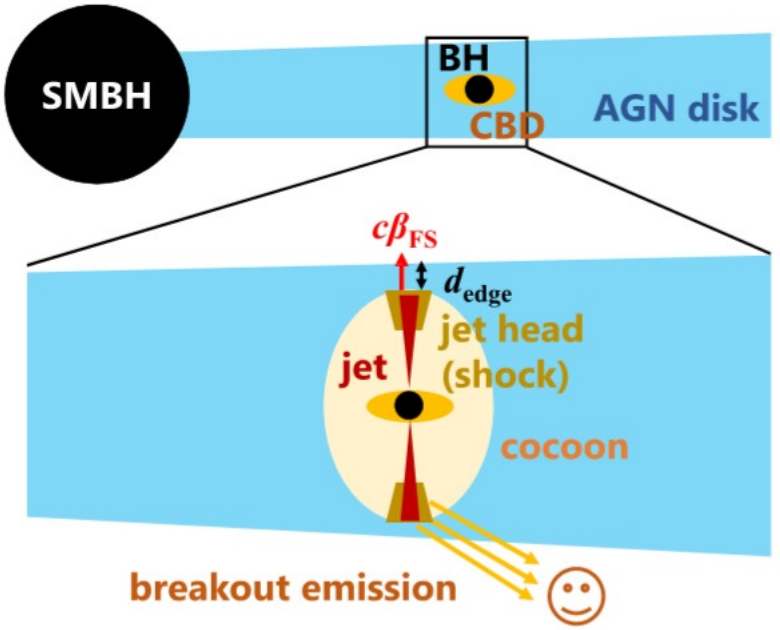
- Hierarchical merging in Active Galactic Nuclei (AGN) accretion disks solves the mass gap problem.
- BHs in AGN sink toward center of disk, form binaries, and rapidly merge.
- High mass remnant remains in AGN, allows hierarchical merger to large masses (Tagawa+ 2021).



Tagawa+ 2020

Mergers in AGN could produce EM counterpart

- Several other formation scenarios, dense star clusters, PBHs, etc.
- Merger's in AGN are particularly interesting due to gas rich environment, possibility of EM counterparts (Tagawa+ 2023) and multi-messenger astronomy with BBH mergers.
- Already claimed coincidences between BBH mergers and AGN flares (Graham+ 2020), but alternative proof of connection is needed – e.g. localisation.

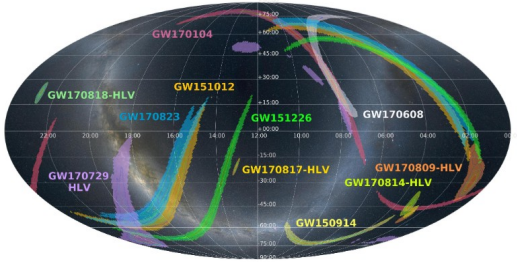


Tagawa+ 2023

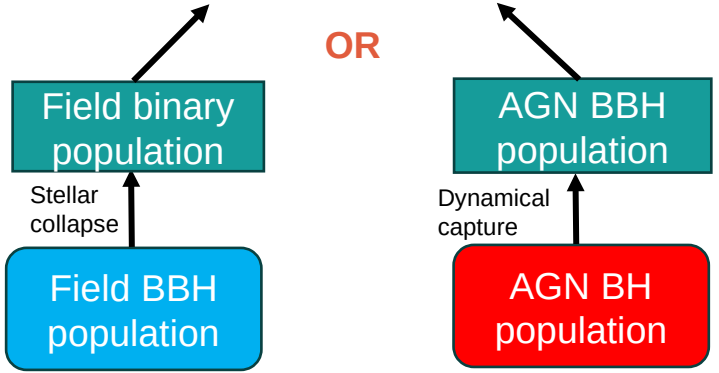
Localisation via the angular cross correlation

Goal:

- Determine if BBH mergers occur primarily in AGN, or non-AGN galaxies, from GW sky localisations.



GW skymap, Abbott+ 2018



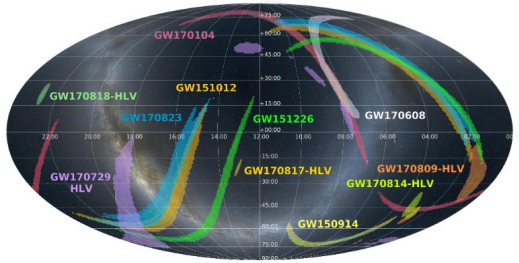
Localisation via the angular cross correlation

Goal:

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Method:

- Angular cross correlation of GW skymaps with galaxy catalogs (catalog of AGN, and catalog of non-AGN galaxies)
- Measures (angular) clustering of skymaps (AGN cluster differently to non-AGN, Banerjee+ 2023)
- Cross correlating shows common clustering (if relative bias b=1), with much lower error if one catalog is more complete.



GW skymap, Abbott+ 2018

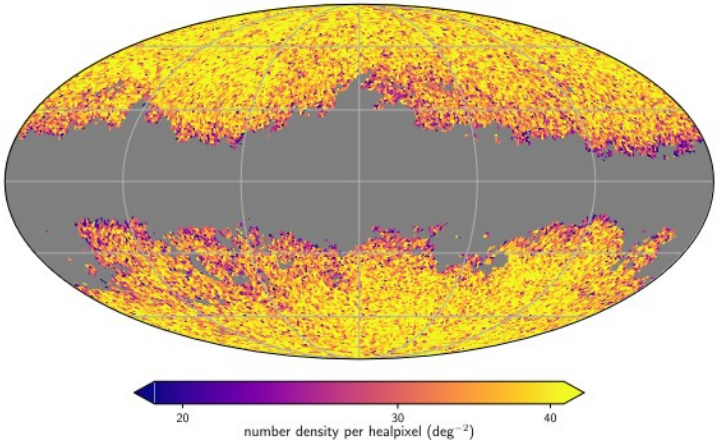
$$C_\ell^{X,Y} = b \cdot C_\ell^{X,X}, \text{where} \quad X = Y \implies b = 1.$$

Angular cross correlation — Bias — Auto correlation

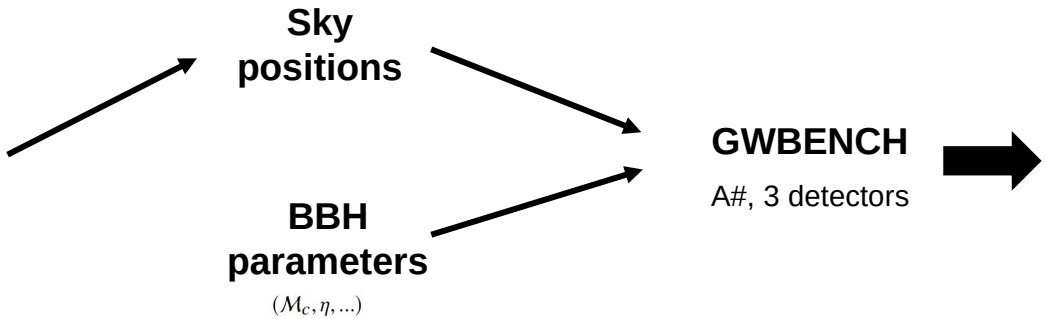
Generating sample sky maps

Issue:

- Too few BBH mergers have been detected to make statistical association with a particular galaxy type, and most localisations are to poorly localised to show clustering information (universe is uniform on angular scales $\Delta\Omega \gg 10\mathrm{deg}^2$).
- Simulate method on realistic samples of future GW sky maps, projecting when we will be able to confidently associate BBH population with either AGN or non-AGN galaxies, using the method of angular cross correlation.



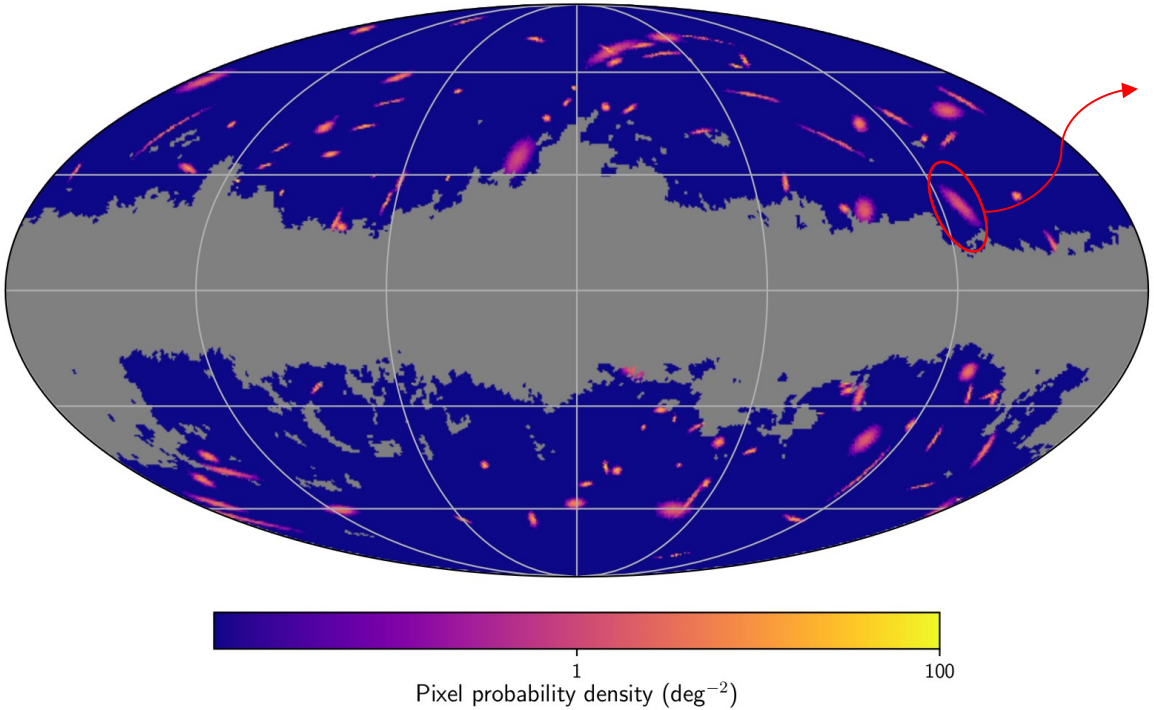
Quaia AGN catalog (Storey-Fisher+ 2024)
GLADE+ all sky catalog (Dálya+ 2021)



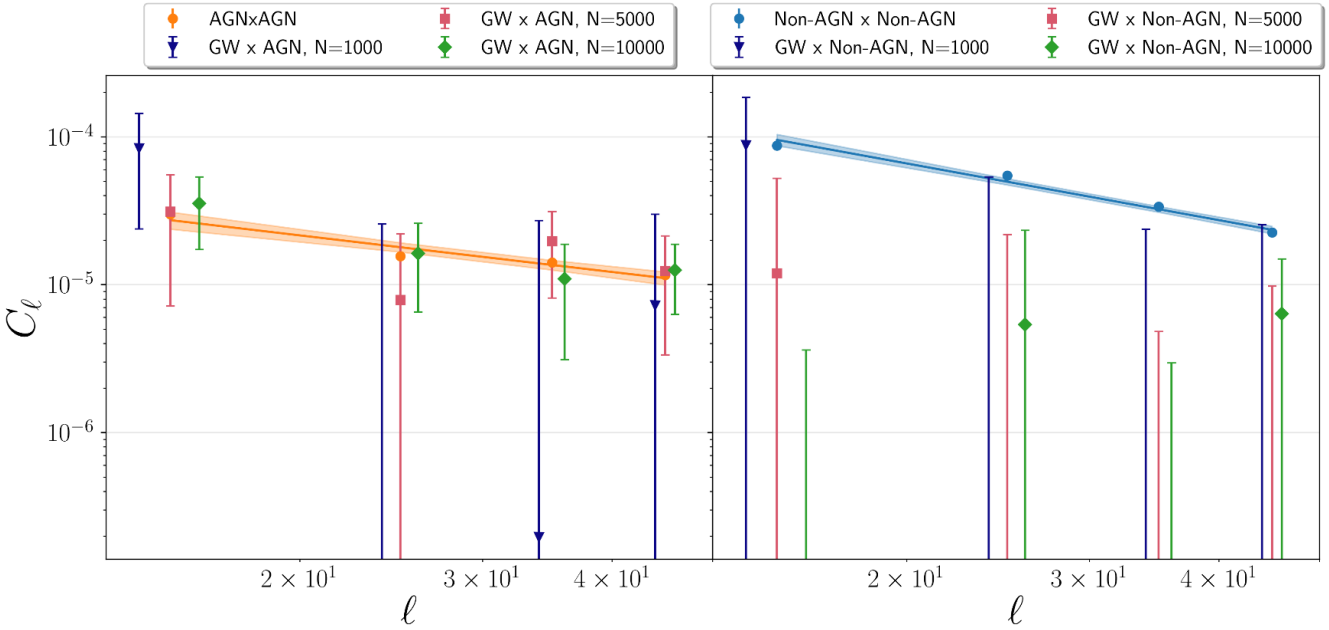
Generating sample sky maps

GWBENCH

A#, 3 detectors



Identifying GW origin

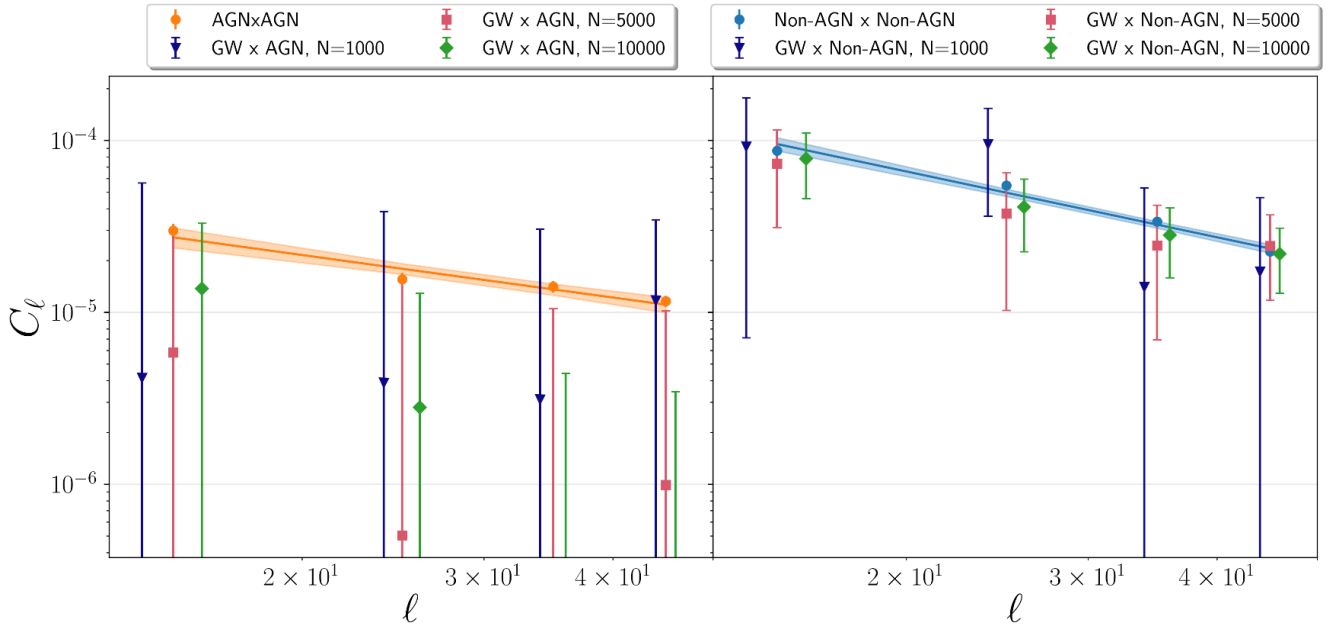


$C_\ell^{GW,AGN} \rightarrow C_\ell^{AGN,AGN}$

~~$C_\ell^{GW,N} \rightarrow C_\ell^{N,N}$~~

AGN origin

Identifying GW origin



$C_\ell^{GW,AGN} \not\rightarrow C_\ell^{AGN,AGN}$

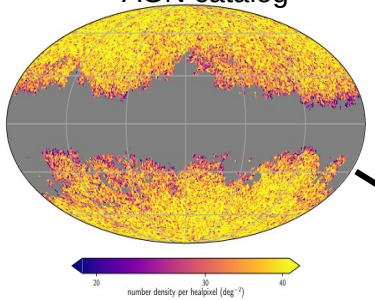
$C_\ell^{GW,N} \rightarrow C_\ell^{N,N}$

➡ Non-AGN origin

Need ~5,000 detections

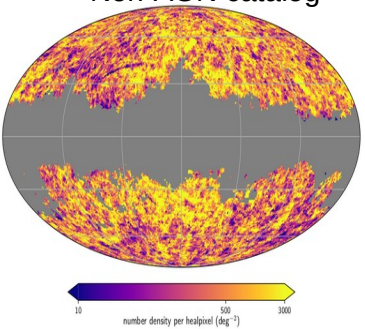
What if we have a mixture?

AGN catalog

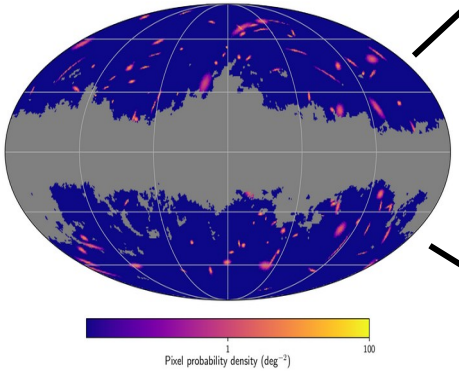


f_A

Non-AGN catalog



f_N



Mixed GW skymap

Cross correlate with AGN catalog → $C_\ell^{A,GW}$

Cross correlate with non-AGN catalog → $C_\ell^{N,GW}$

What if we have a mixture?

- Break ACCs into contribution from each source: $$C_\ell^{X,Y+Z} = f_Y \cdot C_\ell^{X,Y} + f_Z \cdot C_\ell^{X,Z}$$

$$C_\ell^{A,GW} \sim f_A \cdot C_\ell^{A,AGW} + f_N \cdot C_\ell^{A,NGW} + (1 - f_A - f_N) \cdot \epsilon^A$$

AGN contribution

Non-AGN contribution

$$C_\ell^{N,GW} \sim f_N \cdot C_\ell^{N,NGW} + f_A \cdot C_\ell^{N,AGW} + (1 - f_A - f_N) \cdot \epsilon^N$$

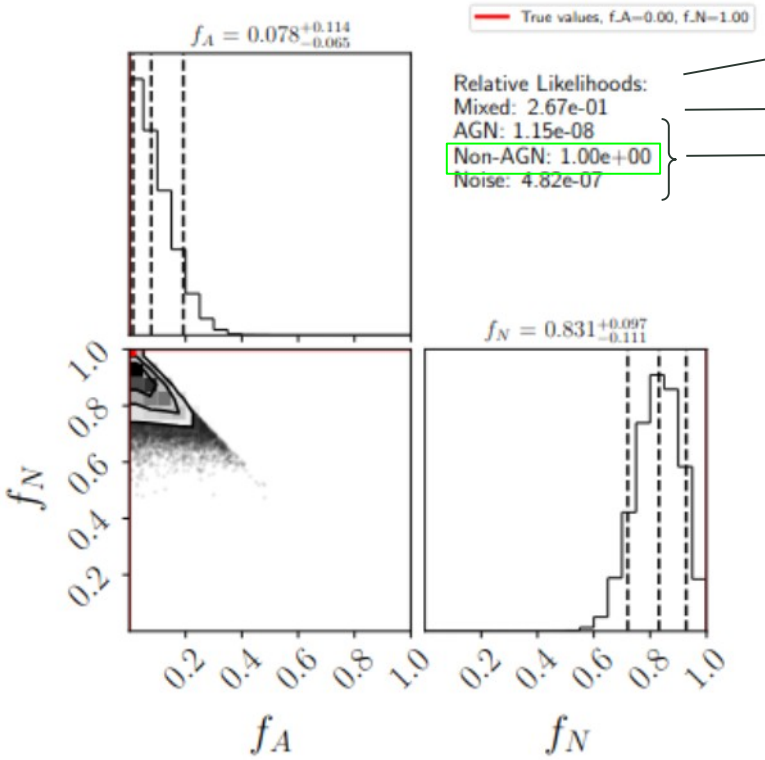
Non-AGN contribution

AGN contribution

Noise terms

- Each GW contribution is well modelled by a Gaussian likelihood, so using uniform priors on fractions we can use Bayes theorem to get posteriors on fractional contributions

All non-AGN catalog case



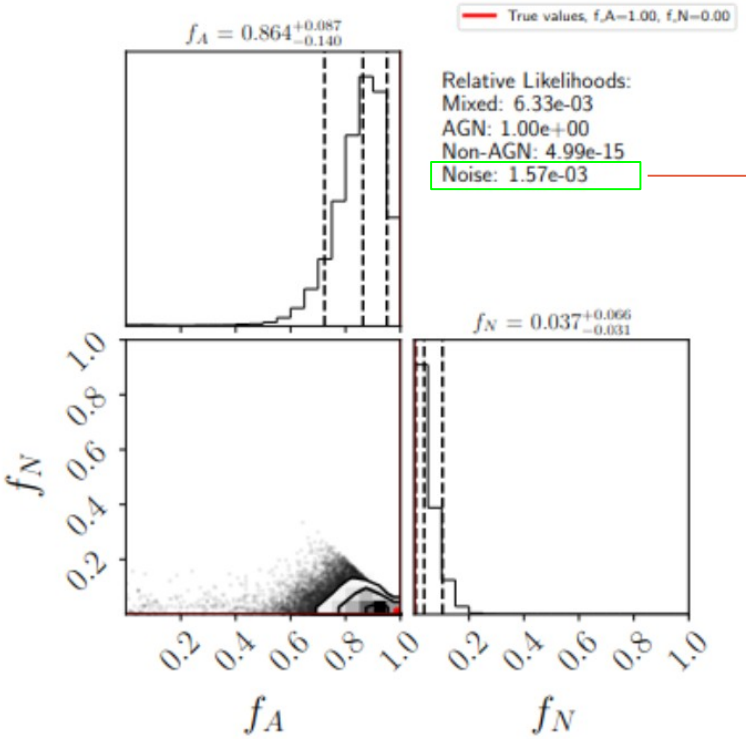
$$\exp\left((\mathrm{AIC}_{\min} - \mathrm{AIC}_i)/2\right)$$

k=2 free parameters

k=1 free parameters

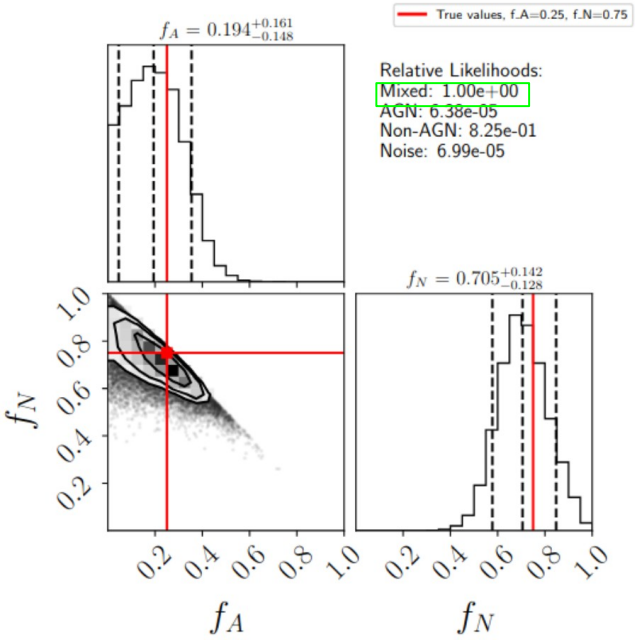
$$\mathrm{AIC} = 2k - \log \hat{L}$$

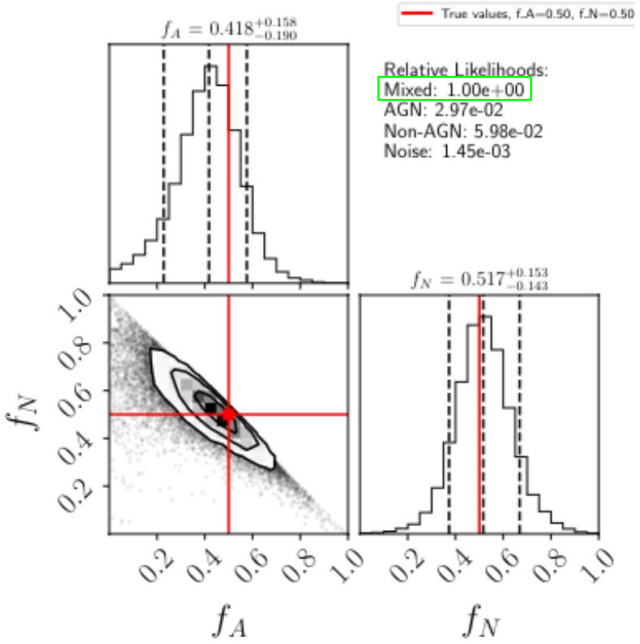
All AGN catalog case

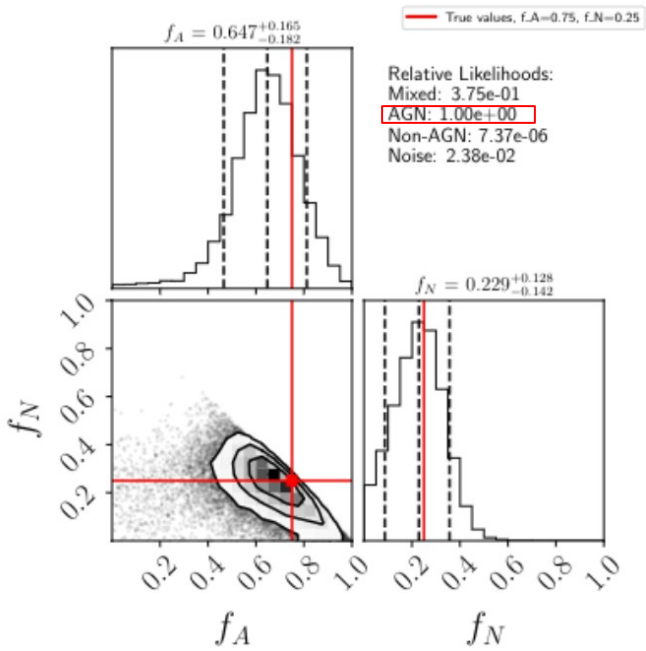


Higher noise, less complete AGN catalog

Mixed catalog case

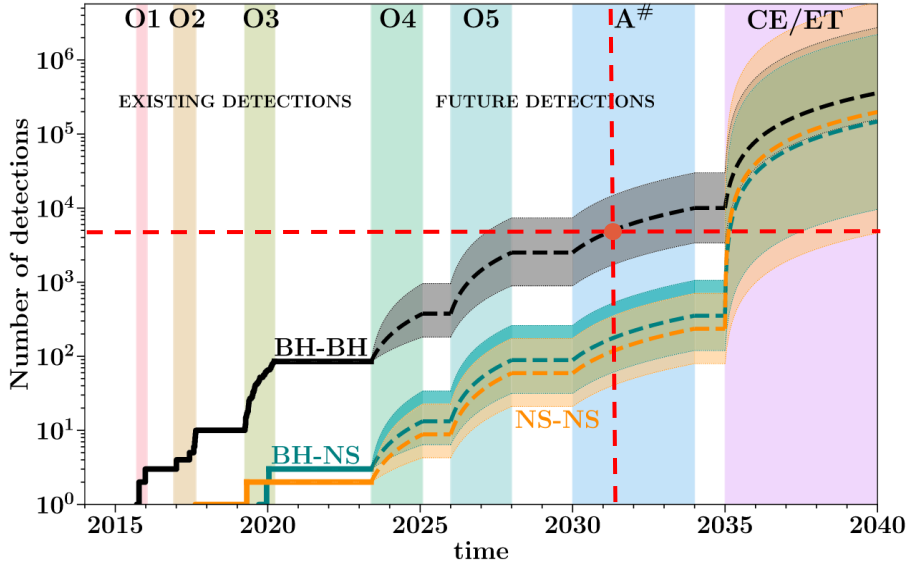






Summary

- BBH mergers in AGN are very interesting, but determining if this channel is dominant is very challenging.
- We have developed a formalism for determining the AGN fraction of BBH mergers in LVK data, using the method of ACCs.
- Requires ~5,000 detections at A# sensitivity.



Broekgaarden+ 2024

Jordan W N Moncrieff, Fiona H Panther, Connecting the hierarchically merging binary black hole population to their host galaxies, Monthly Notices of the Royal Astronomical Society, Volume 543, Issue 2, October 2025, Pages 1833–1841, https://doi.org/10.1093/mnras/staf1582

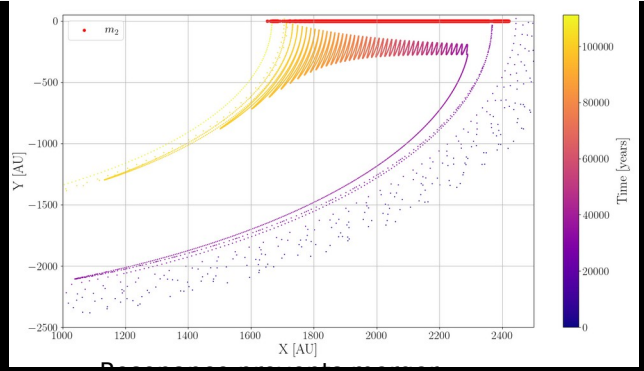
Email: jordan.moncrieff@research.uwa.edu.au

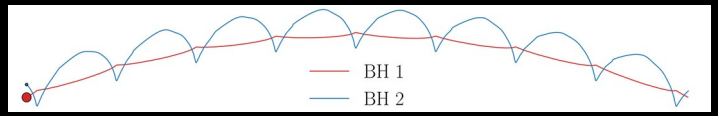
Supermassive black hole

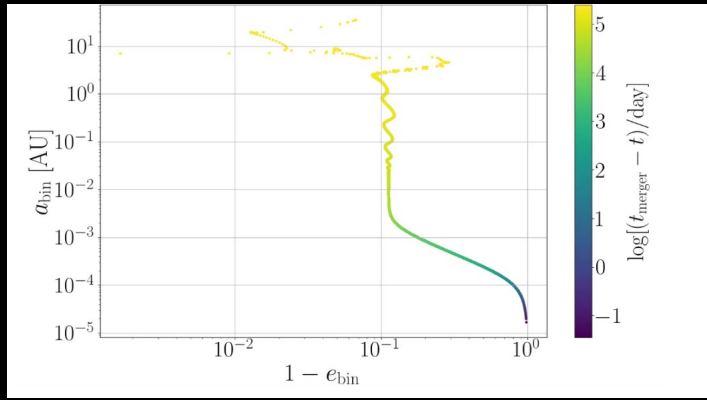
AGN disk

Stellar mass black holes

High mass ratio encounter in disk







=> Limits on the AGN capable of producing high mass ratio events like GW190814